Java Programming – Simple Practice Program

Fasustena S (2548313)

Data Types and Variable.

```
A. import java.util.Scanner;
B.
C. public class P1a {
D.
E. static int globalVar = 100; // Global variable
F.
G. public static void main(String[] args) {
H.
I. Scanner sc = new Scanner(System.in);
J.
K. System.out.println(" 1. Variable Declaration ");
L. int x = 25;
M. double y = 55000.50;
N. char z = 'A';
O. boolean w = true;
P. String v = "Tina";
Q.
R. System.out.println("int: " + x);
S. System.out.println("double: " + y);
T. System.out.println("char: " + z);
U. System.out.println("boolean: " + w);
V. System.out.println("String: " + v);
W.
X.
Y. System.out.println("\n 2: Swaping ");
Z. System.out.print("Enter first number: ");
AA.int a = sc.nextInt();
BB.System.out.print("Enter second number: ");
CC.int b = sc.nextInt();
DD.
EE.System.out.println("Before swap: a = " + a + ", b = " + b);
FF.
GG.a = a + b;
HH.b = a - b;
II.a = a - b;
JJ.
KK.System.out.println("After swap: a = " + a + ", b = " + b);
LL.
MM.
NN.System.out.println("\n 3. Personal Information ");
OO.System.out.print("Enter your age: ");
PP.int Age = sc.nextInt();
QQ.
RR.System.out.print("Enter your height (in feet): ");
```

```
SS.double Height = sc.nextDouble();
TT.
UU.sc.nextLine();
VV.System.out.print("Enter your name: ");
WW.String Name = sc.nextLine();
XX.
YY.System.out.println("\nPerson Details:");
ZZ.System.out.println("Name: " + Name);
AAA.     System.out.println("Age: " + Age + " years");
BBB.     System.out.println("Height: " + Height + " feet");
CCC.
DDD.
EEE.     System.out.println("\n 4. Rectangle Area ");
FFF.     System.out.print("Enter length of rectangle: ");
GGG.     double length = sc.nextDouble();
HHH.
III.     System.out.print("Enter width of rectangle: ");
JJJ.     double width = sc.nextDouble();
KKK.
LLL.     double area = length * width;
MMM.
NNN.     System.out.println("Length: " + length);
OOO.     System.out.println("Width: " + width);
PPP.     System.out.println("Area: " + area);
QQQ.
RRR.
SSS.     System.out.println("\n 5: Multiple Variable Declaration ");
TTT.     int m = 5, n = 10, o = 15;
UUU.     double p = 1.1, q = 2.2, r = 3.3;
VVV.
WWW.     System.out.println("m = " + m + ", n = " + n + ", o = " + o);
XXX.     System.out.println("p = " + p + ", q = " + q + ", r = " + r);
YYY.
ZZZ.
AAAA.    System.out.println("\n  6: Int vs Double ");
BBBB.    int intResult = 10 / 3;
CCCC.    double doubleResult = 10.0 / 3.0;
DDDD.
EEEE.    System.out.println("int division (10 / 3): " + intResult);
FFFF.    System.out.println("double division (10.0 / 3.0): " + doubleResult);
GGGG.
HHHH.
IIII.    System.out.println("\n   7. ASCII Value ");
JJJJ.    System.out.print("Enter a character: ");
KKKK.    char ch = sc.next().charAt(0);
LLLL.    int asciiValue = ch;
MMMM.
NNNN.    System.out.println("Character: " + ch);
OOOO.    System.out.println("ASCII Value: " + asciiValue);
PPPP.
```

```
QQQQ.
RRRR.
SSSS.     System.out.println("\n 8. Simple Interest ");
TTTT.     System.out.print("Enter principal amount: ");
UUUU.     double principal = sc.nextDouble();
VVVV.
WWWW.     System.out.print("Enter rate of interest (%): ");
XXXX.     double rate = sc.nextDouble();
YYYY.
ZZZZ.     System.out.print("Enter time period (years): ");
AAAAA.    double time = sc.nextDouble();
BBBBB.
CCCCC.    double simpleInterest = (principal * rate * time) / 100;
DDDDD.
EEEEE.    System.out.println("\nCalculation Results:");
FFFFF.    System.out.println("Principal: $" + principal);
GGGGG.    System.out.println("Rate: " + rate + "%");
HHHHH.    System.out.println("Time: " + time + " years");
IIIII.    System.out.println("Simple Interest: $" + simpleInterest);
JJJJJ.    System.out.println("Total Amount: $" + (principal + simpleInterest));
KKKKK.
LLLLL.
MMMMM.
NNNNN.    System.out.println("\n 9: Final Variable ");
OOOOO.    final double PI = 3.14159;
PPPPP.    System.out.print("Enter radius of circle: ");
QQQQQ.    double radius = sc.nextDouble();
RRRRR.    double circleArea = PI * radius * radius;
SSSSS.
TTTTT.    System.out.println("PI (constant): " + PI);
UUUUU.    System.out.println("Radius: " + radius);
VVVVV.    System.out.println("Circle Area: " + circleArea);
WWWWW.
XXXXX.    System.out.println("\n 10: Variable Scope ");
YYYYY.    int localVar = 50;  // Local variable
ZZZZZ.
AAAAAA.   System.out.println("Global variable: " + globalVar);
BBBBBB.   System.out.println("Local variable: " + localVar);
CCCCCC.
DDDDDD.
EEEEEE.
FFFFFF.   sc.close();
GGGGGG.
HHHHHH.   }
```

Operators and Expressions

```
IIIIIII.  }
JJJJJJJ.  import java.util.Scanner;
KKKKKK.
LLLLLL.  public class P1b {
MMMMMM.
NNNNNN.      public static void main(String[] args) {
OOOOOO.
PPPPPP.          Scanner scanner = new Scanner(System.in);
QQQQQQ.
RRRRRR.
SSSSSS.          System.out.println(" 1: Arithmetic Operators ");
TTTTTT.          System.out.print("Enter first number: ");
UUUUUU.          int num1 = scanner.nextInt();
VVVVVV.          System.out.print("Enter second number: ");
WWWWWW.          int num2 = scanner.nextInt();
XXXXXX.
YYYYYY.          System.out.println("\nArithmetic Operations:");
ZZZZZZ.          System.out.println(num1 + " + " + num2 + " = " + (num1 + num2));
AAAAAAA.         System.out.println(num1 + " - " + num2 + " = " + (num1 - num2));
BBBBBBB.         System.out.println(num1 + " * " + num2 + " = " + (num1 * num2));
CCCCCCC.         System.out.println(num1 + " / " + num2 + " = " + (num1 / num2));
DDDDDDD.         System.out.println(num1 + " % " + num2 + " = " + (num1 % num2));
EEEEEEE.
FFFFFFF.
GGGGGGG.
HHHHHHH.         System.out.println("\n  2: Even or Odd Check ");
IIIIIII.         System.out.print("Enter a number: ");
JJJJJJJ.         int number = scanner.nextInt();
KKKKKKK.
LLLLLLL.         String result = (number % 2 == 0) ? "Even" : "Odd";
MMMMMMM.         System.out.println(number + " is " + result);
NNNNNNN.
OOOOOOO.
PPPPPPP.         System.out.println("\n  3: Relational Operators ");
QQQQQQQ.         System.out.print("Enter first number: ");
RRRRRRR.         int a = scanner.nextInt();
SSSSSSS.         System.out.print("Enter second number: ");
TTTTTTT.         int b = scanner.nextInt();
UUUUUUU.
VVVVVVV.         System.out.println("\nComparison Results:");
WWWWWWW.         System.out.println(a + " == " + b + " : " + (a == b));
XXXXXXX.         System.out.println(a + " != " + b + " : " + (a != b));
YYYYYYY.         System.out.println(a + " > " + b + " : " + (a > b));
ZZZZZZZ.         System.out.println(a + " < " + b + " : " + (a < b));
AAAAAAAA.        System.out.println(a + " >= " + b + " : " + (a >= b));
BBBBBBBB.        System.out.println(a + " <= " + b + " : " + (a <= b));
CCCCCCCC.
DDDDDDDD.        System.out.println("\n  4: Voting Eligibility (Logical Operators) ");
```

```
EEEEEEEE.        System.out.print("Enter your age: ");
FFFFFFFF.        int age = scanner.nextInt();
GGGGGGGG.        System.out.print("Are you a citizen? (true/false): ");
HHHHHHHH.        boolean isCitizen = scanner.nextBoolean();
IIIIIIII.
JJJJJJJJ.        boolean canVote = (age >= 18) && isCitizen;
KKKKKKKK.
LLLLLLLL.        System.out.println("\nEligibility Check:");
MMMMMMMM.        System.out.println("Age >= 18: " + (age >= 18));
NNNNNNNN.        System.out.println("Is Citizen: " + isCitizen);
OOOOOOOO.        System.out.println("Can Vote (AND): " + canVote);
PPPPPPPP.
QQQQQQQQ.
RRRRRRRR.
SSSSSSSS.
TTTTTTTT.        System.out.println("\n Task 5: Maximum of Two Numbers ");
UUUUUUUU.        System.out.print("Enter first number: ");
VVVVVVVV.        int x = scanner.nextInt();
WWWWWWWW.        System.out.print("Enter second number: ");
XXXXXXXX.        int y = scanner.nextInt();
YYYYYYYY.
ZZZZZZZZ.        int max = (x > y) ? x : y;
AAAAAAAAA.             System.out.println("Maximum of " + x + " and " + y + " is: " +
   max);
BBBBBBBBB.
CCCCCCCCC.
DDDDDDDDD.             System.out.println("\n 6: Increment and Decrement Operators ");
EEEEEEEEE.             System.out.print("Enter a number: ");
FFFFFFFFF.             int value = scanner.nextInt();
GGGGGGGGG.
HHHHHHHHH.             System.out.println("\nOriginal value: " + value);
IIIIIIIII.
JJJJJJJJJ.             System.out.println("Post-increment (value++): " + (value++));
KKKKKKKKK.             System.out.println("After post-increment: " + value);
LLLLLLLLL.
MMMMMMMMM.             System.out.println("Pre-increment (++value): " + (++value));
NNNNNNNNN.             System.out.println("After pre-increment: " + value);
OOOOOOOOO.
PPPPPPPPP.             System.out.println("Post-decrement (value--): " + (value--));
QQQQQQQQQ.             System.out.println("After post-decrement: " + value);
RRRRRRRRR.
SSSSSSSSS.             System.out.println("Pre-decrement (--value): " + (--value));
TTTTTTTTT.             System.out.println("After pre-decrement: " + value);
UUUUUUUUU.
VVVVVVVVV.
WWWWWWWWW.             System.out.println("\n 7: Complex Expression ");
XXXXXXXXX.             System.out.print("Enter value for p: ");
YYYYYYYYY.             int p = scanner.nextInt();
ZZZZZZZZZ.             System.out.print("Enter value for q: ");
AAAAAAAAAA.            int q = scanner.nextInt();
```

```
BBBBBBBBBB.             System.out.print("Enter value for r: ");
CCCCCCCCCC.             int r = scanner.nextInt();
DDDDDDDDDD.
EEEEEEEEEE.
FFFFFFFFFF.             int complexResult = (p + q) * r - p / q + r % p;
GGGGGGGGGG.
HHHHHHHHHH.             System.out.println("\nExpression: (p + q) * r - p / q + r % p");
IIIIIIIIII.             System.out.println("Step by step:");
JJJJJJJJJJ.             System.out.println("(p + q) = (" + p + " + " + q + ") = " + (p +
   q));
KKKKKKKKKK.             System.out.println("(p + q) * r = " + (p + q) + " * " + r + " = "
   + ((p + q) * r));
LLLLLLLLLL.             System.out.println("p / q = " + p + " / " + q + " = " + (p / q));
MMMMMMMMMM.             System.out.println("r % p = " + r + " % " + p + " = " + (r % p));
NNNNNNNNNN.             System.out.println("Final Result: " + complexResult);
OOOOOOOOOO.
PPPPPPPPPP.
QQQQQQQQQQ.
RRRRRRRRRR.             System.out.println("\n 8: Leap Year Check ");
SSSSSSSSSS.             System.out.print("Enter a year: ");
TTTTTTTTTT.             int year = scanner.nextInt();
UUUUUUUUUU.
VVVVVVVVVV.
WWWWWWWWWW.             boolean isLeapYear = (year % 4 == 0 && year % 100 != 0) || (year %
   400 == 0);
XXXXXXXXXX.
YYYYYYYYYY.             System.out.println("\nLeap Year Analysis:");
ZZZZZZZZZZ.             System.out.println("Divisible by 4: " + (year % 4 == 0));
AAAAAAAAAAA.            System.out.println("Divisible by 100: " + (year % 100 == 0));
BBBBBBBBBBB.            System.out.println("Divisible by 400: " + (year % 400 == 0));
CCCCCCCCCCC.            System.out.println(year + " is " + (isLeapYear ? "a LEAP year" :
   "NOT a leap year"));
DDDDDDDDDDD.
EEEEEEEEEEE.
FFFFFFFFFFF.
GGGGGGGGGGG.            System.out.println("\n 9: Operator Precedence ");
HHHHHHHHHHH.            System.out.print("Enter value for m: ");
IIIIIIIIIII.            int m = scanner.nextInt();
JJJJJJJJJJJ.            System.out.print("Enter value for n: ");
KKKKKKKKKKK.            int n = scanner.nextInt();
LLLLLLLLLLL.
MMMMMMMMMMM.            // Demonstrating precedence: multiplication before addition
NNNNNNNNNNN.            int precedenceResult1 = m + n * 2;
OOOOOOOOOOO.            int precedenceResult2 = (m + n) * 2;
PPPPPPPPPPP.
QQQQQQQQQQQ.            System.out.println("\nPrecedence Demonstration:");
RRRRRRRRRRR.            System.out.println("Expression 1: m + n * 2");
SSSSSSSSSSS.            System.out.println("Result: " + m + " + " + n + " * 2 = " +
   precedenceResult1);
TTTTTTTTTTT.            System.out.println("(Multiplication happens first)");
```

```
UUUUUUUUUUU.
VVVVVVVVVVV.            System.out.println("\nExpression 2: (m + n) * 2");
WWWWWWWWWWW.            System.out.println("Result: (" + m + " + " + n + ") * 2 = " +
   precedenceResult2);
XXXXXXXXXXX.            System.out.println("(Parentheses override precedence)");
YYYYYYYYYYY.
ZZZZZZZZZZZ.
AAAAAAAAAAAA.
BBBBBBBBBBBB.           System.out.println("\n 10: Assignment Operators ");
CCCCCCCCCCCC.           System.out.print("Enter initial value: ");
DDDDDDDDDDDD.           int val = scanner.nextInt();
EEEEEEEEEEEE.           System.out.print("Enter operand value: ");
FFFFFFFFFFFF.           int operand = scanner.nextInt();
GGGGGGGGGGGG.
HHHHHHHHHHHH.           int temp = val;
IIIIIIIIIIII.           System.out.println("\nInitial value: " + temp);
JJJJJJJJJJJJ.
KKKKKKKKKKKK.           // Addition assignment
LLLLLLLLLLLL.           temp += operand;
MMMMMMMMMMMM.           System.out.println("After += " + operand + ": " + temp);
NNNNNNNNNNNN.
OOOOOOOOOOOO.           // Subtraction assignment
PPPPPPPPPPPP.           temp -= operand;
QQQQQQQQQQQQ.           System.out.println("After -= " + operand + ": " + temp);
RRRRRRRRRRRR.
SSSSSSSSSSSS.           // Multiplication assignment
TTTTTTTTTTTT.           temp *= operand;
UUUUUUUUUUUU.           System.out.println("After *= " + operand + ": " + temp);
VVVVVVVVVVVV.
WWWWWWWWWWWW.           // Division assignment
XXXXXXXXXXXX.           temp /= operand;
YYYYYYYYYYYY.           System.out.println("After /= " + operand + ": " + temp);
ZZZZZZZZZZZZ.
AAAAAAAAAAAA.           // Modulus assignment
BBBBBBBBBBBB.           temp %= operand;
CCCCCCCCCCCC.           System.out.println("After %= " + operand + ": " + temp);
DDDDDDDDDDDD.
EEEEEEEEEEEE.           // All operators in one demonstration
FFFFFFFFFFFF.           System.out.println("\nAll Assignment Operators Summary:");
GGGGGGGGGGGG.           int demo = val;
HHHHHHHHHHHH.           System.out.println("Starting value: " + demo);
IIIIIIIIIIII.           demo += 5;
JJJJJJJJJJJJ.           System.out.println("demo += 5: " + demo);
KKKKKKKKKKKK.           demo -= 2;
LLLLLLLLLLLL.           System.out.println("demo -= 2: " + demo);
MMMMMMMMMMMM.           demo *= 3;
NNNNNNNNNNNN.           System.out.println("demo *= 3: " + demo);
OOOOOOOOOOOO.           demo /= 4;
PPPPPPPPPPPP.           System.out.println("demo /= 4: " + demo);
QQQQQQQQQQQQ.           demo %= 5;
```

```
RRRRRRRRRRRRR.          System.out.println("demo %= 5: " + demo);
SSSSSSSSSSSSS.
TTTTTTTTTTTTT.
UUUUUUUUUUUUU.
VVVVVVVVVVVVV.          scanner.close();
WWWWWWWWWWWWW.      }
XXXXXXXXXXXXX.  }
```

Type Conversion and Casting

```
YYYYYYYYYYYYY.  import java.util.Scanner;
ZZZZZZZZZZZZZ.
AAAAAAAAAAAAAA. public class P1c {
BBBBBBBBBBBBBB.
CCCCCCCCCCCCCC.     public static void main(String[] args) {
DDDDDDDDDDDDDD.
EEEEEEEEEEEEEE.         Scanner scanner = new Scanner(System.in);
FFFFFFFFFFFFFF.
GGGGGGGGGGGGGG.
HHHHHHHHHHHHHH.         System.out.println(" 1: Implicit Conversion (int to double) ");
IIIIIIIIIIIIII.         System.out.print("Enter an integer: ");
JJJJJJJJJJJJJJ.         int intValue = scanner.nextInt();
KKKKKKKKKKKKKK.
LLLLLLLLLLLLLL.         double doubleValue = intValue;  // Implicit conversion (widening)
MMMMMMMMMMMMMM.
NNNNNNNNNNNNNN.         System.out.println("Integer value: " + intValue);
OOOOOOOOOOOOOO.         System.out.println("Double value (after implicit conversion): " +
   doubleValue);
PPPPPPPPPPPPPP.         System.out.println("Type: int -> double (No data loss)");
QQQQQQQQQQQQQQ.
RRRRRRRRRRRRRR.
SSSSSSSSSSSSSS.         System.out.println("\n 2: Explicit Casting (double to int) ");
TTTTTTTTTTTTTT.         System.out.print("Enter a decimal number: ");
UUUUUUUUUUUUUU.         double decimalValue = scanner.nextDouble();
VVVVVVVVVVVVVV.
WWWWWWWWWWWWWW.         int castedInt = (int) decimalValue;  // Explicit casting
   (narrowing)
XXXXXXXXXXXXXX.
YYYYYYYYYYYYYY.         System.out.println("Original double value: " + decimalValue);
ZZZZZZZZZZZZZZ.         System.out.println("Casted to int: " + castedInt);
AAAAAAAAAAAAAAA.              System.out.println("Type: double -> int (Decimal part
   lost)");
BBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCC.              System.out.println("\n 3: Convert char to ASCII ");
DDDDDDDDDDDDDDD.              System.out.print("Enter a character: ");
EEEEEEEEEEEEEEE.              char character = scanner.next().charAt(0);
FFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGG.              int asciiValue = character;  // Implicit conversion
HHHHHHHHHHHHHHH.              int explicitAscii = (int) character;  // Explicit conversion
IIIIIIIIIIIIIIII.
```

```
JJJJJJJJJJJJJJJ.            System.out.println("Character: " + character);
KKKKKKKKKKKKKKK.            System.out.println("ASCII value (implicit): " + asciiValue);
LLLLLLLLLLLLLLL.            System.out.println("ASCII value (explicit): " +
   explicitAscii);
MMMMMMMMMMMMMMM.
NNNNNNNNNNNNNNN.            // Converting ASCII back to char
OOOOOOOOOOOOOOO.            char backToChar = (char) asciiValue;
PPPPPPPPPPPPPPP.            System.out.println("ASCII " + asciiValue + " back to char: "
   + backToChar);
QQQQQQQQQQQQQQQ.
RRRRRRRRRRRRRRR.
SSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTT.            System.out.println("\n 4: Fix Integer Division Using Casting
   ");
UUUUUUUUUUUUUUU.            System.out.print("Enter dividend: ");
VVVVVVVVVVVVVVV.            int dividend = scanner.nextInt();
WWWWWWWWWWWWWWW.            System.out.print("Enter divisor: ");
XXXXXXXXXXXXXXX.            int divisor = scanner.nextInt();
YYYYYYYYYYYYYYY.
ZZZZZZZZZZZZZZZ.            // Integer division (loses decimal part)
AAAAAAAAAAAAAAAA.           int intDivision = dividend / divisor;
BBBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCCC.           // Fixed using casting to double
DDDDDDDDDDDDDDDD.           double fixedDivision1 = (double) dividend / divisor;
EEEEEEEEEEEEEEEE.           double fixedDivision2 = dividend / (double) divisor;
FFFFFFFFFFFFFFFF.           double fixedDivision3 = (double) (dividend) / (double)
   (divisor);
GGGGGGGGGGGGGGGG.
HHHHHHHHHHHHHHHH.           System.out.println("\nInteger division: " + dividend + " / "
   + divisor + " = " + intDivision);
IIIIIIIIIIIIIIII.           System.out.println("Fixed division (cast dividend): " +
   fixedDivision1);
JJJJJJJJJJJJJJJJ.           System.out.println("Fixed division (cast divisor): " +
   fixedDivision2);
KKKKKKKKKKKKKKKK.           System.out.println("Fixed division (cast both): " +
   fixedDivision3);
LLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMM.
NNNNNNNNNNNNNNNN.
OOOOOOOOOOOOOOOO.           System.out.println("\n 5: Convert long to int ");
PPPPPPPPPPPPPPPP.           System.out.print("Enter a long value: ");
QQQQQQQQQQQQQQQQ.           long longValue = scanner.nextLong();
RRRRRRRRRRRRRRRR.
SSSSSSSSSSSSSSSS.           int intFromLong = (int) longValue;  // Explicit casting
   required
TTTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUUU.           System.out.println("Original long value: " + longValue);
VVVVVVVVVVVVVVVV.           System.out.println("Converted to int: " + intFromLong);
WWWWWWWWWWWWWWWW.           System.out.println("Range of int: " + Integer.MIN_VALUE + "
   to " + Integer.MAX_VALUE);
```

```
XXXXXXXXXXXXXXXX.
YYYYYYYYYYYYYYYY.            if (longValue > Integer.MAX_VALUE || longValue <
   Integer.MIN_VALUE) {
ZZZZZZZZZZZZZZZZ.                System.err.println("WARNING: Data overflow occurred!");
AAAAAAAAAAAAAAAAA.           } else {
BBBBBBBBBBBBBBBBB.               System.err.println("No data loss in this conversion.");
CCCCCCCCCCCCCCCCC.           }
DDDDDDDDDDDDDDDDD.
EEEEEEEEEEEEEEEEE.
FFFFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGGGG.           System.out.println("\n 6: Byte Promotion ");
HHHHHHHHHHHHHHHHH.           System.out.print("Enter first byte value (-128 to 127): ");
IIIIIIIIIIIIIIIII.           byte byte1 = scanner.nextByte();
JJJJJJJJJJJJJJJJJ.           System.out.print("Enter second byte value (-128 to 127): ");
KKKKKKKKKKKKKKKKK.           byte byte2 = scanner.nextByte();
LLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMM.           // Bytes are promoted to int in expressions
NNNNNNNNNNNNNNNNN.           int sum = byte1 + byte2;  // byte + byte = int (promotion)
OOOOOOOOOOOOOOOOO.
PPPPPPPPPPPPPPPPP.           System.out.println("\nByte 1: " + byte1);
QQQQQQQQQQQQQQQQQ.           System.out.println("Byte 2: " + byte2);
RRRRRRRRRRRRRRRRR.           System.out.println("Sum (automatically promoted to int): " +
   sum);
SSSSSSSSSSSSSSSSS.           System.out.println("Type of result: int");
TTTTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUUUU.           // To store back in byte, explicit casting needed
VVVVVVVVVVVVVVVVV.           byte byteSum = (byte) (byte1 + byte2);
WWWWWWWWWWWWWWWWW.           System.out.println("Sum casted back to byte: " + byteSum);
XXXXXXXXXXXXXXXXX.
YYYYYYYYYYYYYYYYY.           // Demonstrating overflow
ZZZZZZZZZZZZZZZZZ.           byte b1 = 100;
AAAAAAAAAAAAAAAAA.           byte b2 = 50;
BBBBBBBBBBBBBBBBB.           byte overflowResult = (byte) (b1 + b2);
CCCCCCCCCCCCCCCCC.           System.out.println("\nOverflow example: 100 + 50 = " + (b1 +
   b2) + " (int)");
DDDDDDDDDDDDDDDDD.           System.out.println("Casted to byte: " + overflowResult + "
   (overflow occurred)");
EEEEEEEEEEEEEEEEE.
FFFFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGGGG.
HHHHHHHHHHHHHHHHH.           System.out.println("\n 7: Convert String to Integer ");
IIIIIIIIIIIIIIIII.           scanner.nextLine();  // Consume newline
JJJJJJJJJJJJJJJJJ.           System.out.print("Enter a number as string: ");
KKKKKKKKKKKKKKKKK.           String numberString = scanner.nextLine();
LLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMM.           try {
NNNNNNNNNNNNNNNNN.               int parsedInt = Integer.parseInt(numberString);
OOOOOOOOOOOOOOOOO.               double parsedDouble = Double.parseDouble(numberString);
PPPPPPPPPPPPPPPPP.
```

```
QQQQQQQQQQQQQQQQQQ.                System.out.println("String value: \"" + numberString +
    "\"");
RRRRRRRRRRRRRRRRRR.                System.out.println("Parsed to int: " + parsedInt);
SSSSSSSSSSSSSSSSSS.                System.out.println("Parsed to double: " + parsedDouble);
TTTTTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUUUUU.                // String to other types
VVVVVVVVVVVVVVVVVV.                System.out.print("\nEnter a decimal number as string:
    ");
WWWWWWWWWWWWWWWWWW.                String decimalString = scanner.nextLine();
XXXXXXXXXXXXXXXXXX.                double decimal = Double.parseDouble(decimalString);
YYYYYYYYYYYYYYYYYY.                float floatValue = Float.parseFloat(decimalString);
ZZZZZZZZZZZZZZZZZZ.
AAAAAAAAAAAAAAAAAA.                System.out.println("String: \"" + decimalString + "\"");
BBBBBBBBBBBBBBBBBB.                System.out.println("To double: " + decimal);
CCCCCCCCCCCCCCCCCC.                System.out.println("To float: " + floatValue);
DDDDDDDDDDDDDDDDDD.
EEEEEEEEEEEEEEEEEE.            } catch (NumberFormatException e) {
FFFFFFFFFFFFFFFFFF.                System.err.println("Error: Invalid number format!");
GGGGGGGGGGGGGGGGGG.            }
HHHHHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIIIII.            System.out.println("\n 8: Data Loss During Casting ");
JJJJJJJJJJJJJJJJJJ.            System.out.print("Enter a large double value: ");
KKKKKKKKKKKKKKKKKK.            double largeDouble = scanner.nextDouble();
LLLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMMM.            int lossyInt = (int) largeDouble;
NNNNNNNNNNNNNNNNNN.            byte lossyByte = (byte) largeDouble;
OOOOOOOOOOOOOOOOOO.
PPPPPPPPPPPPPPPPPP.            System.out.println("\nOriginal double: " + largeDouble);
QQQQQQQQQQQQQQQQQQ.            System.out.println("Casted to int: " + lossyInt + " (decimal
    lost)");
RRRRRRRRRRRRRRRRRR.            System.out.println("Casted to byte: " + lossyByte + "
    (severe data loss)");
SSSSSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTTTTT.            // Demonstrating precision loss
UUUUUUUUUUUUUUUUUU.            double preciseValue = 123.456789;
VVVVVVVVVVVVVVVVVV.            float lessPresice = (float) preciseValue;
WWWWWWWWWWWWWWWWWW.
XXXXXXXXXXXXXXXXXX.            System.out.println("\nPrecision loss example:");
YYYYYYYYYYYYYYYYYY.            System.out.println("Double: " + preciseValue);
ZZZZZZZZZZZZZZZZZZ.            System.out.println("Float: " + lessPresice);
AAAAAAAAAAAAAAAAAA.
BBBBBBBBBBBBBBBBBB.            // Large to small type
CCCCCCCCCCCCCCCCCC.            long largeLong = 2147483648L;  // Greater than int max
DDDDDDDDDDDDDDDDDD.            int overflowInt = (int) largeLong;
EEEEEEEEEEEEEEEEEE.
FFFFFFFFFFFFFFFFFF.            System.out.println("\nOverflow example:");
GGGGGGGGGGGGGGGGGG.            System.out.println("Long value: " + largeLong);
HHHHHHHHHHHHHHHHHH.            System.out.println("Int max: " + Integer.MAX_VALUE);
IIIIIIIIIIIIIIIIII.            System.out.println("Casted to int: " + overflowInt + "
    (overflow!)");
```

```
JJJJJJJJJJJJJJJJJJJJ.
KKKKKKKKKKKKKKKKKKKK.
LLLLLLLLLLLLLLLLLLLL.        System.out.println("\n 9: Temperature Conversion Using
   Casting ");
MMMMMMMMMMMMMMMMMMMM.        System.out.print("Enter temperature in Fahrenheit: ");
NNNNNNNNNNNNNNNNNNNN.        double fahrenheit = scanner.nextDouble();
OOOOOOOOOOOOOOOOOOOO.
PPPPPPPPPPPPPPPPPPPP.        // Celsius = (Fahrenheit - 32) * 5/9
QQQQQQQQQQQQQQQQQQQQ.        // Without casting - integer division problem
RRRRRRRRRRRRRRRRRRRR.        double celsiusWrong = (fahrenheit - 32) * 5 / 9;  // 5/9 = 0
   in integer
SSSSSSSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTTTTTTT.        // With casting - correct conversion
UUUUUUUUUUUUUUUUUUUU.        double celsiusCorrect = (fahrenheit - 32) * 5.0 / 9.0;
VVVVVVVVVVVVVVVVVVVV.        double celsiusCasted = ((fahrenheit - 32) * (double) 5) /
   (double) 9;
WWWWWWWWWWWWWWWWWWWW.
XXXXXXXXXXXXXXXXXXXX.        System.out.println("\nFahrenheit: " + fahrenheit + "°F");
YYYYYYYYYYYYYYYYYYYY.        System.out.println("Wrong conversion (5/9 as int): " +
   celsiusWrong + "°C");
ZZZZZZZZZZZZZZZZZZZZ.        System.out.println("Correct conversion (5.0/9.0): " +
   celsiusCorrect + "°C");
AAAAAAAAAAAAAAAAAAAA.              System.out.println("Correct conversion (with casting):
   " + celsiusCasted + "°C");
BBBBBBBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCCCCCCC.              // Reverse conversion
DDDDDDDDDDDDDDDDDDDD.              System.out.print("\nEnter temperature in Celsius: ");
EEEEEEEEEEEEEEEEEEEE.              double celsius = scanner.nextDouble();
FFFFFFFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGGGGGGG.              // Fahrenheit = (Celsius * 9/5) + 32
HHHHHHHHHHHHHHHHHHHH.              double fahrenheitResult = (celsius * 9.0 / 5.0) + 32;
IIIIIIIIIIIIIIIIIIII.
JJJJJJJJJJJJJJJJJJJJ.              System.out.println("Celsius: " + celsius + "°C");
KKKKKKKKKKKKKKKKKKKK.              System.out.println("Fahrenheit: " + fahrenheitResult +
   "°F");
LLLLLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMMMMM.
NNNNNNNNNNNNNNNNNNNN.
OOOOOOOOOOOOOOOOOOOO.              System.out.println("\n 10: Automatic Type Promotion in
   Expressions ");
PPPPPPPPPPPPPPPPPPPP.              System.out.print("Enter a byte value: ");
QQQQQQQQQQQQQQQQQQQQ.              byte byteVal = scanner.nextByte();
RRRRRRRRRRRRRRRRRRRR.              System.out.print("Enter a short value: ");
SSSSSSSSSSSSSSSSSSSS.              short shortVal = scanner.nextShort();
TTTTTTTTTTTTTTTTTTTT.              System.out.print("Enter an int value: ");
UUUUUUUUUUUUUUUUUUUU.              int intVal = scanner.nextInt();
VVVVVVVVVVVVVVVVVVVV.              System.out.print("Enter a long value: ");
WWWWWWWWWWWWWWWWWWWW.              long longVal = scanner.nextLong();
XXXXXXXXXXXXXXXXXXXX.              System.out.print("Enter a float value: ");
YYYYYYYYYYYYYYYYYYYY.              float floatVal = scanner.nextFloat();
```

```
ZZZZZZZZZZZZZZZZZZZZZ.              System.out.print("Enter a double value: ");
AAAAAAAAAAAAAAAAAAAAAA.             double doubleVal = scanner.nextDouble();
BBBBBBBBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCCCCCCCC.              System.out.println("\nType Promotion in Mixed
   Expressions:");
DDDDDDDDDDDDDDDDDDDDDD.
EEEEEEEEEEEEEEEEEEEEE.              // byte + short -> int
FFFFFFFFFFFFFFFFFFFFF.              int result1 = byteVal + shortVal;
GGGGGGGGGGGGGGGGGGGGG.              System.out.println("byte + short = int: " + byteVal +
   " + " + shortVal + " = " + result1);
HHHHHHHHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIIIIIIII.              // int + long -> long
JJJJJJJJJJJJJJJJJJJJJ.              long result2 = intVal + longVal;
KKKKKKKKKKKKKKKKKKKKK.              System.out.println("int + long = long: " + intVal + "
   + " + longVal + " = " + result2);
LLLLLLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMMMMMM.              // long + float -> float
NNNNNNNNNNNNNNNNNNNNN.              float result3 = longVal + floatVal;
OOOOOOOOOOOOOOOOOOOOO.              System.out.println("long + float = float: " + longVal
   + " + " + floatVal + " = " + result3);
PPPPPPPPPPPPPPPPPPPPP.
QQQQQQQQQQQQQQQQQQQQQ.              // float + double -> double
RRRRRRRRRRRRRRRRRRRRR.              double result4 = floatVal + doubleVal;
SSSSSSSSSSSSSSSSSSSSS.              System.out.println("float + double = double: " +
   floatVal + " + " + doubleVal + " = " + result4);
TTTTTTTTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUUUUUUUU.              // Complex expression with multiple types
VVVVVVVVVVVVVVVVVVVVV.              double complexResult = byteVal + shortVal * intVal -
   longVal / floatVal + doubleVal;
WWWWWWWWWWWWWWWWWWWWW.              System.out.println("\nComplex expression (all
   types):");
XXXXXXXXXXXXXXXXXXXXX.              System.out.println("byte + short * int - long / float
   + double");
YYYYYYYYYYYYYYYYYYYYY.              System.out.println("Result (promoted to double): " +
   complexResult);
ZZZZZZZZZZZZZZZZZZZZZ.
AAAAAAAAAAAAAAAAAAAAAA.             // Promotion hierarchy demonstration
BBBBBBBBBBBBBBBBBBBBB.              System.out.println("\nType Promotion Hierarchy:");
CCCCCCCCCCCCCCCCCCCCC.              System.out.println("byte -> short -> int -> long ->
   float -> double");
DDDDDDDDDDDDDDDDDDDDDD.             System.out.println("(Lower types automatically promote
   to higher types in expressions)");
EEEEEEEEEEEEEEEEEEEEE.
FFFFFFFFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGGGGGGGG.
HHHHHHHHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIIIIIIII.              scanner.close();
JJJJJJJJJJJJJJJJJJJJJ.          }
KKKKKKKKKKKKKKKKKKKKK.      }
```

Standard Input / Output

```
LLLLLLLLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMMMMMMMM.
NNNNNNNNNNNNNNNNNNNNNNN.
OOOOOOOOOOOOOOOOOOOOOOO.
PPPPPPPPPPPPPPPPPPPPPPP.
QQQQQQQQQQQQQQQQQQQQQQQ.
RRRRRRRRRRRRRRRRRRRRRRR.
SSSSSSSSSSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUUUUUUUUUU.
VVVVVVVVVVVVVVVVVVVVVVV.
WWWWWWWWWWWWWWWWWWWWWWW.
XXXXXXXXXXXXXXXXXXXXXXX.
YYYYYYYYYYYYYYYYYYYYYYY.
ZZZZZZZZZZZZZZZZZZZZZZZ.
AAAAAAAAAAAAAAAAAAAAAAA.
BBBBBBBBBBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCCCCCCCCCC.
DDDDDDDDDDDDDDDDDDDDDDD.
EEEEEEEEEEEEEEEEEEEEEEE.
FFFFFFFFFFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGGGGGGGGGG.
HHHHHHHHHHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIIIIIIIIIII.
JJJJJJJJJJJJJJJJJJJJJJJJ.
KKKKKKKKKKKKKKKKKKKKKKK.
LLLLLLLLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMMMMMMMM.
NNNNNNNNNNNNNNNNNNNNNNN.
OOOOOOOOOOOOOOOOOOOOOOO.
PPPPPPPPPPPPPPPPPPPPPPP.
QQQQQQQQQQQQQQQQQQQQQQQ.
RRRRRRRRRRRRRRRRRRRRRRR.
SSSSSSSSSSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUUUUUUUUUU.
VVVVVVVVVVVVVVVVVVVVVVV.
WWWWWWWWWWWWWWWWWWWWWWW.
```

```java
import java.util.Scanner;

public class P1d {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println(" 1: Read Name and Age ");
        System.out.print("Enter your name: ");
        String name = scanner.nextLine();
        System.out.print("Enter your age: ");
        int age = scanner.nextInt();

        System.out.println("\nOutput:");
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Welcome, " + name + "! You are " + age + " years old.");



        System.out.println("\n 2: Read Two Numbers and Print Sum ");
        System.out.print("Enter first number: ");
        int num1 = scanner.nextInt();
        System.out.print("Enter second number: ");
        int num2 = scanner.nextInt();

        int sum = num1 + num2;

        System.out.println("\nOutput:");
        System.out.println("First Number: " + num1);
        System.out.println("Second Number: " + num2);
        System.out.println("Sum: " + num1 + " + " + num2 + " = " + sum);



        System.out.println("\n 3: Read Float and Print Square ");
        System.out.print("Enter a decimal number: ");
```

```
XXXXXXXXXXXXXXXXXXXXXXXX.          float number = scanner.nextFloat();
YYYYYYYYYYYYYYYYYYYYYYYY.
ZZZZZZZZZZZZZZZZZZZZZZZZ.          float square = number * number;
AAAAAAAAAAAAAAAAAAAAAAAA.
BBBBBBBBBBBBBBBBBBBBBBBB.          System.out.println("\nOutput:");
CCCCCCCCCCCCCCCCCCCCCCCC.          System.out.println("Number: " + number);
DDDDDDDDDDDDDDDDDDDDDDDD.          System.out.println("Square: " + number + " × " +
   number + " = " + square);
EEEEEEEEEEEEEEEEEEEEEEEE.          System.out.printf("Square (formatted): %.2f%n",
   square);
FFFFFFFFFFFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGGGGGGGGGGG.
HHHHHHHHHHHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIIIIIIIIIIII.         System.out.println("\n 4: Read a Character ");
JJJJJJJJJJJJJJJJJJJJJJJJ.          System.out.print("Enter a single character: ");
KKKKKKKKKKKKKKKKKKKKKKKK.          char character = scanner.next().charAt(0);
LLLLLLLLLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMMMMMMMMM.          System.out.println("\nOutput:");
NNNNNNNNNNNNNNNNNNNNNNNN.          System.out.println("You entered: " + character);
OOOOOOOOOOOOOOOOOOOOOOOO.          System.out.println("ASCII value: " + (int)character);
PPPPPPPPPPPPPPPPPPPPPPPP.          System.out.println("Is letter: " +
   Character.isLetter(character));
QQQQQQQQQQQQQQQQQQQQQQQQ.          System.out.println("Is digit: " +
   Character.isDigit(character));
RRRRRRRRRRRRRRRRRRRRRRRR.
SSSSSSSSSSSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTTTTTTTTTTT.          System.out.println("\n 5: Read and Print Multiple
   Inputs ");
UUUUUUUUUUUUUUUUUUUUUUUU.          scanner.nextLine(); // Consume newline
VVVVVVVVVVVVVVVVVVVVVVVV.
WWWWWWWWWWWWWWWWWWWWWWWW.          System.out.print("Enter your full name: ");
XXXXXXXXXXXXXXXXXXXXXXXX.          String fullName = scanner.nextLine();
YYYYYYYYYYYYYYYYYYYYYYYY.
ZZZZZZZZZZZZZZZZZZZZZZZZ.          System.out.print("Enter your city: ");
AAAAAAAAAAAAAAAAAAAAAAAA.          String city = scanner.nextLine();
BBBBBBBBBBBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCCCCCCCCCCC.          System.out.print("Enter your age: ");
DDDDDDDDDDDDDDDDDDDDDDDD.          int personAge = scanner.nextInt();
EEEEEEEEEEEEEEEEEEEEEEEE.
FFFFFFFFFFFFFFFFFFFFFFFF.          System.out.print("Enter your height (in meters): ");
GGGGGGGGGGGGGGGGGGGGGGGG.          double height = scanner.nextDouble();
HHHHHHHHHHHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIIIIIIIIIIII.         System.out.print("Are you employed? (true/false): ");
JJJJJJJJJJJJJJJJJJJJJJJJ.          boolean isEmployed = scanner.nextBoolean();
KKKKKKKKKKKKKKKKKKKKKKKK.
LLLLLLLLLLLLLLLLLLLLLLLL.          System.out.println("\n--- Personal Information ---");
MMMMMMMMMMMMMMMMMMMMMMMM.          System.out.println("Full Name: " + fullName);
NNNNNNNNNNNNNNNNNNNNNNNN.          System.out.println("City: " + city);
OOOOOOOOOOOOOOOOOOOOOOOO.          System.out.println("Age: " + personAge);
PPPPPPPPPPPPPPPPPPPPPPPP.          System.out.println("Height: " + height + " meters");
```

```
QQQQQQQQQQQQQQQQQQQQQQQQQQ.         System.out.println("Employed: " + isEmployed);
RRRRRRRRRRRRRRRRRRRRRRRRRR.
SSSSSSSSSSSSSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUUUUUUUUUUUUU.         System.out.println("\n 6: Difference between next()
   and nextLine() ");
VVVVVVVVVVVVVVVVVVVVVVVVVV.         scanner.nextLine(); // Consume newline
WWWWWWWWWWWWWWWWWWWWWWWWWW.
XXXXXXXXXXXXXXXXXXXXXXXXXX.         System.out.println("Using next() - reads single
   word:");
YYYYYYYYYYYYYYYYYYYYYYYYYY.         System.out.print("Enter your first name: ");
ZZZZZZZZZZZZZZZZZZZZZZZZZZ.         String firstName = scanner.next();
AAAAAAAAAAAAAAAAAAAAAAAAAA.         System.out.println("You entered: " + firstName);
BBBBBBBBBBBBBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCCCCCCCCCCCCC.         scanner.nextLine(); // Consume remaining line
DDDDDDDDDDDDDDDDDDDDDDDDDD.
EEEEEEEEEEEEEEEEEEEEEEEEEE.         System.out.println("\nUsing nextLine() - reads entire
   line:");
FFFFFFFFFFFFFFFFFFFFFFFFFF.         System.out.print("Enter your full sentence: ");
GGGGGGGGGGGGGGGGGGGGGGGGGG.         String sentence = scanner.nextLine();
HHHHHHHHHHHHHHHHHHHHHHHHHH.         System.out.println("You entered: " + sentence);
IIIIIIIIIIIIIIIIIIIIIIIIII.
JJJJJJJJJJJJJJJJJJJJJJJJJJ.         System.out.println("\nDemonstration:");
KKKKKKKKKKKKKKKKKKKKKKKKKK.         System.out.print("Type multiple words and press Enter:
   ");
LLLLLLLLLLLLLLLLLLLLLLLLLL.         String word = scanner.next();
MMMMMMMMMMMMMMMMMMMMMMMMMM.         String restOfLine = scanner.nextLine();
NNNNNNNNNNNNNNNNNNNNNNNNNN.
OOOOOOOOOOOOOOOOOOOOOOOOOO.         System.out.println("next() captured: \"" + word +
   "\"");
PPPPPPPPPPPPPPPPPPPPPPPPPP.         System.out.println("Remaining line: \"" + restOfLine +
   "\"");
QQQQQQQQQQQQQQQQQQQQQQQQQQ.
RRRRRRRRRRRRRRRRRRRRRRRRRR.         System.out.println("\nKey Differences:");
SSSSSSSSSSSSSSSSSSSSSSSSSS.         System.out.println("next()     - Reads until
   whitespace (space, tab, newline)");
TTTTTTTTTTTTTTTTTTTTTTTTTT.         System.out.println("nextLine() - Reads entire line
   until Enter key");
UUUUUUUUUUUUUUUUUUUUUUUUUU.
VVVVVVVVVVVVVVVVVVVVVVVVVV.
WWWWWWWWWWWWWWWWWWWWWWWWWW.
XXXXXXXXXXXXXXXXXXXXXXXXXX.         System.out.println("\n 7: Calculate Average Marks ");
YYYYYYYYYYYYYYYYYYYYYYYYYY.         System.out.print("Enter number of subjects: ");
ZZZZZZZZZZZZZZZZZZZZZZZZZZ.         int numSubjects = scanner.nextInt();
AAAAAAAAAAAAAAAAAAAAAAAAAA.
BBBBBBBBBBBBBBBBBBBBBBBBBB.               int totalMarks = 0;
CCCCCCCCCCCCCCCCCCCCCCCCCC.
DDDDDDDDDDDDDDDDDDDDDDDDDD.               System.out.println("\nEnter marks for each
   subject:");
EEEEEEEEEEEEEEEEEEEEEEEEEE.               for (int i = 1; i <= numSubjects; i++) {
```

```
FFFFFFFFFFFFFFFFFFFFFFFFFFFF.                System.out.print("Subject " + i + ": ");
GGGGGGGGGGGGGGGGGGGGGGGGGGGG.                int marks = scanner.nextInt();
HHHHHHHHHHHHHHHHHHHHHHHHHHHH.                totalMarks += marks;
IIIIIIIIIIIIIIIIIIIIIIIIIIII.            }
JJJJJJJJJJJJJJJJJJJJJJJJJJJJ.
KKKKKKKKKKKKKKKKKKKKKKKKKKKK.            double average = (double) totalMarks /
   numSubjects;
LLLLLLLLLLLLLLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMMMMMMMMMMMMMM.            System.out.println("\n--- Result ---");
NNNNNNNNNNNNNNNNNNNNNNNNNNNN.            System.out.println("Total Marks: " +
   totalMarks);
OOOOOOOOOOOOOOOOOOOOOOOOOOOO.            System.out.println("Number of Subjects: " +
   numSubjects);
PPPPPPPPPPPPPPPPPPPPPPPPPPPP.            System.out.println("Average Marks: " + average);
QQQQQQQQQQQQQQQQQQQQQQQQQQQQ.            System.out.printf("Average (formatted): %.2f%n",
   average);
RRRRRRRRRRRRRRRRRRRRRRRRRRRR.
SSSSSSSSSSSSSSSSSSSSSSSSSSSS.            // Determine grade
TTTTTTTTTTTTTTTTTTTTTTTTTTTT.            String grade;
UUUUUUUUUUUUUUUUUUUUUUUUUUUU.            if (average >= 90) {
VVVVVVVVVVVVVVVVVVVVVVVVVVVV.                grade = "A+";
WWWWWWWWWWWWWWWWWWWWWWWWWWWW.            } else if (average >= 80) {
XXXXXXXXXXXXXXXXXXXXXXXXXXXX.                grade = "A";
YYYYYYYYYYYYYYYYYYYYYYYYYYYY.            } else if (average >= 70) {
ZZZZZZZZZZZZZZZZZZZZZZZZZZZZ.                grade = "B";
AAAAAAAAAAAAAAAAAAAAAAAAAAAA.            } else if (average >= 60) {
BBBBBBBBBBBBBBBBBBBBBBBBBBBB.                grade = "C";
CCCCCCCCCCCCCCCCCCCCCCCCCCCC.            } else if (average >= 50) {
DDDDDDDDDDDDDDDDDDDDDDDDDDDD.                grade = "D";
EEEEEEEEEEEEEEEEEEEEEEEEEEEE.            } else {
FFFFFFFFFFFFFFFFFFFFFFFFFFFF.                grade = "F";
GGGGGGGGGGGGGGGGGGGGGGGGGGGG.            }
HHHHHHHHHHHHHHHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIIIIIIIIIIIIIII.            System.out.println("Grade: " + grade);
JJJJJJJJJJJJJJJJJJJJJJJJJJJJ.
KKKKKKKKKKKKKKKKKKKKKKKKKKKK.
LLLLLLLLLLLLLLLLLLLLLLLLLLLL.            System.out.println("\n 8: Print Error Message
   using System.err");
MMMMMMMMMMMMMMMMMMMMMMMMMMMM.            System.out.print("Enter a number (positive
   only): ");
NNNNNNNNNNNNNNNNNNNNNNNNNNNN.            int userInput = scanner.nextInt();
OOOOOOOOOOOOOOOOOOOOOOOOOOOO.
PPPPPPPPPPPPPPPPPPPPPPPPPPPP.            if (userInput < 0) {
QQQQQQQQQQQQQQQQQQQQQQQQQQQQ.                System.err.println("ERROR: Negative number
   not allowed!");
RRRRRRRRRRRRRRRRRRRRRRRRRRRR.                System.err.println("ERROR: Please enter a
   positive number.");
SSSSSSSSSSSSSSSSSSSSSSSSSSSS.            } else {
TTTTTTTTTTTTTTTTTTTTTTTTTTTT.                System.out.println("SUCCESS: Valid input
   received - " + userInput);
```

```
UUUUUUUUUUUUUUUUUUUUUUUUUUUUU.            }
VVVVVVVVVVVVVVVVVVVVVVVVVVVVV.
WWWWWWWWWWWWWWWWWWWWWWWWWWWWW.            // Demonstrating different output streams
XXXXXXXXXXXXXXXXXXXXXXXXXXXXX.            System.out.println("\nDemonstration of output
   streams:");
YYYYYYYYYYYYYYYYYYYYYYYYYYYYY.            System.out.println("This is printed to standard
   output (System.out)");
ZZZZZZZZZZZZZZZZZZZZZZZZZZZZZ.            System.err.println("This is printed to standard
   error (System.err)");
AAAAAAAAAAAAAAAAAAAAAAAAAAAAA.
BBBBBBBBBBBBBBBBBBBBBBBBBBBBB.            System.out.print("Enter zero to see error, non-
   zero for success: ");
CCCCCCCCCCCCCCCCCCCCCCCCCCCCC.            int testValue = scanner.nextInt();
DDDDDDDDDDDDDDDDDDDDDDDDDDDDD.
EEEEEEEEEEEEEEEEEEEEEEEEEEEEE.            if (testValue == 0) {
FFFFFFFFFFFFFFFFFFFFFFFFFFFFF.                System.err.println("ERROR: Division by zero
   not allowed!");
GGGGGGGGGGGGGGGGGGGGGGGGGGGGG.            } else {
HHHHHHHHHHHHHHHHHHHHHHHHHHHHH.                System.out.println("Result: " + (100 /
   testValue));
IIIIIIIIIIIIIIIIIIIIIIIIIIIII.            }
JJJJJJJJJJJJJJJJJJJJJJJJJJJJJ.
KKKKKKKKKKKKKKKKKKKKKKKKKKKKK.
LLLLLLLLLLLLLLLLLLLLLLLLLLLLL.            // 9. Calculate tax from salary input
MMMMMMMMMMMMMMMMMMMMMMMMMMMMM.            System.out.println("\n 9: Calculate Tax from
   Salary ");
NNNNNNNNNNNNNNNNNNNNNNNNNNNNN.            System.out.print("Enter your annual salary: $");
OOOOOOOOOOOOOOOOOOOOOOOOOOOOO.            double salary = scanner.nextDouble();
PPPPPPPPPPPPPPPPPPPPPPPPPPPPP.
QQQQQQQQQQQQQQQQQQQQQQQQQQQQQ.            double tax = 0;
RRRRRRRRRRRRRRRRRRRRRRRRRRRRR.            double taxRate = 0;
SSSSSSSSSSSSSSSSSSSSSSSSSSSSS.
TTTTTTTTTTTTTTTTTTTTTTTTTTTTT.            // Tax slabs
UUUUUUUUUUUUUUUUUUUUUUUUUUUUU.            if (salary <= 10000) {
VVVVVVVVVVVVVVVVVVVVVVVVVVVVV.                taxRate = 0;
WWWWWWWWWWWWWWWWWWWWWWWWWWWWW.                tax = 0;
XXXXXXXXXXXXXXXXXXXXXXXXXXXXX.            } else if (salary <= 30000) {
YYYYYYYYYYYYYYYYYYYYYYYYYYYYY.                taxRate = 5;
ZZZZZZZZZZZZZZZZZZZZZZZZZZZZZ.                tax = (salary - 10000) * 0.05;
A.         } else if (salary <= 60000) {
B.             taxRate = 10;
C.             tax = (20000 * 0.05) + (salary - 30000) * 0.10;
D.         } else if (salary <= 100000) {
E.             taxRate = 15;
F.             tax = (20000 * 0.05) + (30000 * 0.10) + (salary - 60000) * 0.15;
G.         } else {
H.             taxRate = 20;
I.             tax = (20000 * 0.05) + (30000 * 0.10) + (40000 * 0.15) + (salary - 100000)
   * 0.20;
J.         }
```

```
K.
L.          double netSalary = salary - tax;
M.
N.          System.out.println("\n--- Tax Calculation ---");
O.          System.out.printf("Gross Salary: $%.2f%n", salary);
P.          System.out.printf("Tax Rate: %.0f%%%n", taxRate);
Q.          System.out.printf("Tax Amount: $%.2f%n", tax);
R.          System.out.printf("Net Salary: $%.2f%n", netSalary);
S.
T.          System.out.println("\nTax Slabs:");
U.          System.out.println("Up to $10,000: 0%");
V.          System.out.println("$10,001 - $30,000: 5%");
W.          System.out.println("$30,001 - $60,000: 10%");
X.          System.out.println("$60,001 - $100,000: 15%");
Y.          System.out.println("Above $100,000: 20%");
Z.
AA.
BB.         // 10. Simple login using input
CC.         System.out.println("\n=== Task 10: Simple Login System ===");
DD.         scanner.nextLine(); // Consume newline
EE.
FF.         // Predefined credentials
GG.         final String CORRECT_USERNAME = "admin";
HH.         final String CORRECT_PASSWORD = "password123";
II.
JJ.         System.out.println("--- Login System ---");
KK.         System.out.print("Enter username: ");
LL.         String username = scanner.nextLine();
MM.
NN.         System.out.print("Enter password: ");
OO.         String password = scanner.nextLine();
PP.
QQ.         System.out.println("\nAuthenticating...");
RR.
SS.         if (username.equals(CORRECT_USERNAME) && password.equals(CORRECT_PASSWORD)) {
TT.             System.out.println("✓ Login Successful!");
UU.             System.out.println("Welcome, " + username + "!");
VV.             System.out.println("Access Granted.");
WW.         } else {
XX.             System.err.println("✗ Login Failed!");
YY.
ZZ.             if (!username.equals(CORRECT_USERNAME)) {
AAA.                    System.err.println("ERROR: Invalid username.");
BBB.                }
CCC.                if (!password.equals(CORRECT_PASSWORD)) {
DDD.                    System.err.println("ERROR: Invalid password.");
EEE.                }
FFF.
GGG.                System.err.println("Access Denied.");
HHH.            }
```

```
III.             scanner.close();
JJJ.         }
KKK.     }
```

Command-Line Arguments

```
LLL.
MMM.     public class P1e {
NNN.
OOO.
PPP.         public static void task1_DisplayArguments(String[] args) {
QQQ.             System.out.println("1: Display Command-Line Arguments ");
RRR.
SSS.             if (args.length == 0) {
TTT.                 System.out.println("No arguments provided!");
UUU.             } else {
VVV.                 System.out.println("Number of arguments: " + args.length);
WWW.                 System.out.println("\nArguments received:");
XXX.
YYY.                 for (int i = 0; i < args.length; i++) {
ZZZ.                     System.out.println("args[" + i + "] = " + args[i]);
AAAA.                }
BBBB.            }
CCCC.            System.out.println();
DDDD.        }
EEEE.
FFFF.        public static void task2_AddNumbers(String[] args) {
GGGG.            System.out.println("2: Add Two Numbers Using Arguments ");
HHHH.
IIII.            if (args.length < 2) {
JJJJ.                System.err.println("ERROR: Two numbers required!");
KKKK.                System.out.println("Usage: java CommandLineArgsDemo <num1> <num2>");
LLLL.            } else {
MMMM.                try {
NNNN.                    int num1 = Integer.parseInt(args[0]);
OOOO.                    int num2 = Integer.parseInt(args[1]);
PPPP.                    int sum = num1 + num2;
QQQQ.
RRRR.                    System.out.println("First number: " + num1);
SSSS.                    System.out.println("Second number: " + num2);
TTTT.                    System.out.println("Sum: " + num1 + " + " + num2 + " = " + sum);
UUUU.                } catch (NumberFormatException e) {
VVVV.                    System.err.println("ERROR: Please provide valid numbers!");
WWWW.                }
XXXX.            }
YYYY.            System.out.println();
```

```
ZZZZ.        }
AAAAA.
BBBBB.       public static void task3_MultiplyNumbers(String[] args) {
CCCCC.           System.out.println("3: Multiply Two Numbers ");
DDDDD.
EEEEE.           if (args.length < 2) {
FFFFF.               System.err.println("ERROR: Two numbers required for
   multiplication!");
GGGGG.               System.out.println("Usage: java CommandLineArgsDemo <num1> <num2>");
HHHHH.           } else {
IIIII.               try {
JJJJJ.                   double num1 = Double.parseDouble(args[0]);
KKKKK.                   double num2 = Double.parseDouble(args[1]);
LLLLL.                   double product = num1 * num2;
MMMMM.
NNNNN.                   System.out.println("First number: " + num1);
OOOOO.                   System.out.println("Second number: " + num2);
PPPPP.                   System.out.println("Product: " + num1 + " × " + num2 + " = " +
   product);
QQQQQ.                   System.out.printf("Product (formatted): %.2f%n", product);
RRRRR.               } catch (NumberFormatException e) {
SSSSS.                   System.err.println("ERROR: Please provide valid numbers!");
TTTTT.               }
UUUUU.           }
VVVVV.           System.out.println();
WWWWW.       }
XXXXX.
YYYYY.
ZZZZZ.
AAAAAA.      public static void task4_NameAndAge(String[] args) {
BBBBBB.          System.out.println("4: Print Name and Age via Arguments ");
CCCCCC.
DDDDDD.          if (args.length < 2) {
EEEEEE.              System.err.println("ERROR: Name and age required!");
FFFFFF.              System.out.println("Usage: java CommandLineArgsDemo <name> <age>");
GGGGGG.          } else {
HHHHHH.              try {
IIIIII.                  String name = args[0];
JJJJJJ.                  int age = Integer.parseInt(args[1]);
KKKKKK.
LLLLLL.                  System.out.println("Name: " + name);
MMMMMM.                  System.out.println("Age: " + age);
NNNNNN.                  System.out.println("\nWelcome, " + name + "!");
OOOOOO.                  System.out.println("You are " + age + " years old.");
PPPPPP.
QQQQQQ.              } catch (NumberFormatException e) {
RRRRRR.                  System.err.println("ERROR: Age must be a valid number!");
SSSSSS.          }
TTTTTT.          System.out.println();
UUUUUU.      }
```

```
VVVVVV.
WWWWWW.     }
XXXXXX.     public static void task5_CountArguments(String[] args) {
YYYYYY.         System.out.println("5: Count Command-Line Arguments ");
ZZZZZZ.
AAAAAAA.        int count = args.length;
BBBBBBB.
CCCCCCC.        System.out.println("Total number of arguments: " + count);
DDDDDDD.
EEEEEEE.        if (count == 0) {
FFFFFFF.            System.out.println("No arguments provided.");
GGGGGGG.        } else if (count == 1) {
HHHHHHH.            System.out.println("One argument provided: " + args[0]);
IIIIIII.        } else {
JJJJJJJ.            System.out.println("Multiple arguments provided:");
KKKKKKK.            for (int i = 0; i < count; i++) {
LLLLLLL.                System.out.println("  " + (i + 1) + ". " + args[i]);
MMMMMMM.            }
NNNNNNN.        }
OOOOOOO.
PPPPPPP.        System.out.println("\nArgument Statistics:");
QQQQQQQ.        System.out.println("Total arguments: " + count);
RRRRRRR.        System.out.println("Total characters: " + getTotalCharacters(args));
SSSSSSS.        System.out.println();
TTTTTTT.    }
UUUUUUU.
VVVVVVV.    private static int getTotalCharacters(String[] args) {
WWWWWWW.        int total = 0;
XXXXXXX.        for (String arg : args) {
YYYYYYY.            total += arg.length();
ZZZZZZZ.        }
AAAAAAAA.       return total;
BBBBBBBB.   }
CCCCCCCC.
DDDDDDDD.
EEEEEEEE.   public static void task6_AverageOfThree(String[] args) {
FFFFFFFF.       System.out.println("6: Average of Three Arguments ");
GGGGGGGG.
HHHHHHHH.       if (args.length < 3) {
IIIIIIII.           System.err.println("ERROR: Three numbers required!");
JJJJJJJJ.           System.out.println("Usage: java CommandLineArgsDemo <num1> <num2>
   <num3>");
KKKKKKKK.       } else {
LLLLLLLL.           try {
MMMMMMMM.               double num1 = Double.parseDouble(args[0]);
NNNNNNNN.               double num2 = Double.parseDouble(args[1]);
OOOOOOOO.               double num3 = Double.parseDouble(args[2]);
PPPPPPPP.
QQQQQQQQ.               double sum = num1 + num2 + num3;
RRRRRRRR.               double average = sum / 3;
```

```
SSSSSSSS.
TTTTTTTT.                System.out.println("Number 1: " + num1);
UUUUUUUU.                System.out.println("Number 2: " + num2);
VVVVVVVV.                System.out.println("Number 3: " + num3);
WWWWWWWW.                System.out.println("\nSum: " + sum);
XXXXXXXX.                System.out.println("Average: " + average);
YYYYYYYY.                System.out.printf("Average (formatted): %.2f%n", average);
ZZZZZZZZ.            } catch (NumberFormatException e) {
AAAAAAAAA.                   System.err.println("ERROR: Please provide valid
   numbers!");
BBBBBBBBB.                }
CCCCCCCCC.            }
DDDDDDDDD.            System.out.println();
EEEEEEEEE.        }
FFFFFFFFF.
GGGGGGGGG.
HHHHHHHHH.
IIIIIIIII.        public static void task7_HandleMissingArguments(String[] args) {
JJJJJJJJJ.            System.out.println("7: Handle Missing Arguments ");
KKKKKKKKK.
LLLLLLLLL.            final int REQUIRED_ARGS = 4;
MMMMMMMMM.
NNNNNNNNN.            if (args.length < REQUIRED_ARGS) {
OOOOOOOOO.                System.err.println("ERROR: Insufficient arguments!");
PPPPPPPPP.                System.err.println("Required: " + REQUIRED_ARGS + "
   arguments");
QQQQQQQQQ.                System.err.println("Provided: " + args.length + " arguments");
RRRRRRRRR.                System.out.println("\nUsage: java CommandLineArgsDemo <name>
   <age> <city> <salary>");
SSSSSSSSS.                System.out.println("Example: java CommandLineArgsDemo John 25
   NewYork 50000");
TTTTTTTTT.
UUUUUUUUU.                // Show which arguments are missing
VVVVVVVVV.                System.out.println("\nMissing arguments:");
WWWWWWWWW.                String[] argNames = {"name", "age", "city", "salary"};
XXXXXXXXX.                for (int i = args.length; i < REQUIRED_ARGS; i++) {
YYYYYYYYY.                    System.out.println("  - " + argNames[i]);
ZZZZZZZZZ.                }
AAAAAAAAAA.           } else {
BBBBBBBBBB.               try {
CCCCCCCCCC.                   String name = args[0];
DDDDDDDDDD.                   int age = Integer.parseInt(args[1]);
EEEEEEEEEE.                   String city = args[2];
FFFFFFFFFF.                   double salary = Double.parseDouble(args[3]);
GGGGGGGGGG.
HHHHHHHHHH.                   System.out.println("✓ All arguments received
   successfully!");
IIIIIIIIII.                   System.out.println("\n--- Personal Information ---");
JJJJJJJJJJ.                   System.out.println("Name: " + name);
KKKKKKKKKK.                   System.out.println("Age: " + age);
```

```
LLLLLLLLLL.                     System.out.println("City: " + city);
MMMMMMMMMM.                     System.out.printf("Salary: $%.2f%n", salary);
NNNNNNNNNN.                 } catch (NumberFormatException e) {
OOOOOOOOOO.                     System.err.println("ERROR: Invalid data format!");
PPPPPPPPPP.                     System.err.println("Age and salary must be valid
   numbers.");
QQQQQQQQQQ.                 }
RRRRRRRRRR.             }
SSSSSSSSSS.             System.out.println();
TTTTTTTTTT.         }
UUUUUUUUUU.
VVVVVVVVVV.
WWWWWWWWWW.         public static void task8_FindMaximum(String[] args) {
XXXXXXXXXX.             System.out.println("8: Find Maximum of Two Arguments ");
YYYYYYYYYY.
ZZZZZZZZZZ.             if (args.length < 2) {
AAAAAAAAAAA.                System.err.println("ERROR: Two numbers required!");
BBBBBBBBBBB.                System.out.println("Usage: java CommandLineArgsDemo <num1>
   <num2>");
CCCCCCCCCCC.            } else {
DDDDDDDDDDD.                try {
EEEEEEEEEEE.                    double num1 = Double.parseDouble(args[0]);
FFFFFFFFFFF.                    double num2 = Double.parseDouble(args[1]);
GGGGGGGGGGG.
HHHHHHHHHHH.                    double max = (num1 > num2) ? num1 : num2;
IIIIIIIIIII.                    double min = (num1 < num2) ? num1 : num2;
JJJJJJJJJJJ.
KKKKKKKKKKK.                    System.out.println("First number: " + num1);
LLLLLLLLLLL.                    System.out.println("Second number: " + num2);
MMMMMMMMMMM.                    System.out.println("\nMaximum: " + max);
NNNNNNNNNNN.                    System.out.println("Minimum: " + min);
OOOOOOOOOOO.                    System.out.println("Difference: " + (max - min));
PPPPPPPPPPP.                } catch (NumberFormatException e) {
QQQQQQQQQQQ.                    System.err.println("ERROR: Please provide valid
   numbers!");
RRRRRRRRRRR.                }
SSSSSSSSSSS.            }
TTTTTTTTTTT.            System.out.println();
UUUUUUUUUUU.        }
VVVVVVVVVVV.
WWWWWWWWWWW.
XXXXXXXXXXX.        public static void task9_InterestCalculation(String[] args) {
YYYYYYYYYYY.            System.out.println("9: Interest Calculation Using Arguments ");
ZZZZZZZZZZZ.
AAAAAAAAAAAA.           if (args.length < 3) {
BBBBBBBBBBBB.               System.err.println("ERROR: Three values required!");
CCCCCCCCCCCC.               System.out.println("Usage: java CommandLineArgsDemo
   <principal> <rate> <time>");
DDDDDDDDDDDD.               System.out.println("Example: java CommandLineArgsDemo 10000
   5.5 3");
```

```
EEEEEEEEEEEE.           } else {
FFFFFFFFFFFF.               try {
GGGGGGGGGGGG.                   double principal = Double.parseDouble(args[0]);
HHHHHHHHHHHH.                   double rate = Double.parseDouble(args[1]);
IIIIIIIIIIII.                   double time = Double.parseDouble(args[2]);
JJJJJJJJJJJJ.
KKKKKKKKKKKK.                   // Simple Interest = (P × R × T) / 100
LLLLLLLLLLLL.                   double simpleInterest = (principal * rate * time) / 100;
MMMMMMMMMMMM.
NNNNNNNNNNNN.                   // Compound Interest = P(1 + R/100)^T - P
OOOOOOOOOOOO.                   double compoundInterest = principal * Math.pow((1 + rate /
   100), time) - principal;
PPPPPPPPPPPP.
QQQQQQQQQQQQ.                   System.out.println("--- Input Values ---");
RRRRRRRRRRRR.                   System.out.printf("Principal Amount: $%.2f%n", principal);
SSSSSSSSSSSS.                   System.out.printf("Rate of Interest: %.2f%%%n", rate);
TTTTTTTTTTTT.                   System.out.printf("Time Period: %.2f years%n", time);
UUUUUUUUUUUU.
VVVVVVVVVVVV.                   System.out.println("\n--- Simple Interest ---");
WWWWWWWWWWWW.                   System.out.printf("Interest: $%.2f%n", simpleInterest);
XXXXXXXXXXXX.                   System.out.printf("Total Amount: $%.2f%n", (principal +
   simpleInterest));
YYYYYYYYYYYY.
ZZZZZZZZZZZZ.                   System.out.println("\n--- Compound Interest ---");
AAAAAAAAAAAA.                   System.out.printf("Interest: $%.2f%n", compoundInterest);
BBBBBBBBBBBB.                   System.out.printf("Total Amount: $%.2f%n", (principal +
   compoundInterest));
CCCCCCCCCCCC.
DDDDDDDDDDDD.               } catch (NumberFormatException e) {
EEEEEEEEEEEE.                   System.err.println("ERROR: Please provide valid numeric
   values!");
FFFFFFFFFFFF.               }
GGGGGGGGGGGG.           }
HHHHHHHHHHHH.           System.out.println();
IIIIIIIIIIII.       }
JJJJJJJJJJJJ.
KKKKKKKKKKKK.
LLLLLLLLLLLL.       public static void task10_ReverseArguments(String[] args) {
MMMMMMMMMMMM.           System.out.println("10: Reverse Command-Line Arguments ");
NNNNNNNNNNNN.
OOOOOOOOOOOO.           if (args.length == 0) {
PPPPPPPPPPPP.               System.out.println("No arguments to reverse!");
QQQQQQQQQQQQ.           } else {
RRRRRRRRRRRR.               System.out.println("Original Arguments:");
SSSSSSSSSSSS.               for (int i = 0; i < args.length; i++) {
TTTTTTTTTTTT.                   System.out.println("  args[" + i + "] = " + args[i]);
UUUUUUUUUUUU.               }
VVVVVVVVVVVV.
WWWWWWWWWWWW.               System.out.println("\nReversed Arguments:");
XXXXXXXXXXXX.               for (int i = args.length - 1; i >= 0; i--) {
```

```
YYYYYYYYYYYYY.                System.out.println("  args[" + i + "] = " + args[i]);
ZZZZZZZZZZZZZ.            }
AAAAAAAAAAAAAA.
BBBBBBBBBBBBBB.           // Also reverse each string
CCCCCCCCCCCCCC.           System.out.println("\nReversed Strings:");
DDDDDDDDDDDDDD.           for (int i = 0; i < args.length; i++) {
EEEEEEEEEEEEEE.               String reversed = new
   StringBuilder(args[i]).reverse().toString();
FFFFFFFFFFFFFF.               System.out.println("  " + args[i] + " → " + reversed);
GGGGGGGGGGGGGG.           }
HHHHHHHHHHHHHH.       }
IIIIIIIIIIIIIII.      System.out.println();
JJJJJJJJJJJJJJ.    }
KKKKKKKKKKKKKK.
LLLLLLLLLLLLLL.
MMMMMMMMMMMMMM.    // Main method - runs all tasks with sample data
NNNNNNNNNNNNNN.    public static void main(String[] args) {
OOOOOOOOOOOOOO.
PPPPPPPPPPPPPP.
QQQQQQQQQQQQQQ.       System.out.println("   COMMAND-LINE ARGUMENTS
   DEMONSTRATION        ");
RRRRRRRRRRRRRR.
SSSSSSSSSSSSSS.
TTTTTTTTTTTTTT.       // Check if arguments are provided
UUUUUUUUUUUUUU.       if (args.length == 0) {
VVVVVVVVVVVVVV.           System.out.println("No command-line arguments provided!");
WWWWWWWWWWWWWW.           System.out.println("Running demonstrations with sample
   data...\n");
XXXXXXXXXXXXXX.
YYYYYYYYYYYYYY.           // Sample demonstrations
ZZZZZZZZZZZZZZ.           System.out.println("DEMONSTRATION MODE - Using sample
   data\n");
AAAAAAAAAAAAAAA.                System.out.println("=".repeat(60) + "\n");
BBBBBBBBBBBBBBB.
CCCCCCCCCCCCCCC.                // Task 1
DDDDDDDDDDDDDDD.                task1_DisplayArguments(new String[]{"Hello", "World",
   "Java", "2026"});
EEEEEEEEEEEEEEE.
FFFFFFFFFFFFFFF.                // Task 2
GGGGGGGGGGGGGGG.                task2_AddNumbers(new String[]{"25", "35"});
HHHHHHHHHHHHHHH.
IIIIIIIIIIIIIIII.               // Task 3
JJJJJJJJJJJJJJJ.                task3_MultiplyNumbers(new String[]{"12.5", "4.2"});
KKKKKKKKKKKKKKK.
LLLLLLLLLLLLLLL.                // Task 4
MMMMMMMMMMMMMMM.                task4_NameAndAge(new String[]{"Tina", "28"});
NNNNNNNNNNNNNNN.
OOOOOOOOOOOOOOO.                // Task 5
PPPPPPPPPPPPPPP.                task5_CountArguments(new String[]{"arg1", "arg2",
   "arg3", "arg4", "arg5"});
```

```
QQQQQQQQQQQQQQQ.
RRRRRRRRRRRRRRR.                // Task 6
SSSSSSSSSSSSSSS.                task6_AverageOfThree(new String[]{"85", "90", "78"});
TTTTTTTTTTTTTTT.
UUUUUUUUUUUUUUU.                // Task 7
VVVVVVVVVVVVVVV.                task7_HandleMissingArguments(new String[]{"Anu "});
WWWWWWWWWWWWWWW.
XXXXXXXXXXXXXXX.                // Task 8
YYYYYYYYYYYYYYY.                task8_FindMaximum(new String[]{"45.5", "78.9"});
ZZZZZZZZZZZZZZZ.
AAAAAAAAAAAAAAAA.               // Task 9
BBBBBBBBBBBBBBBB.               task9_InterestCalculation(new String[]{"10000", "7.5",
   "5"});
CCCCCCCCCCCCCCCC.
DDDDDDDDDDDDDDDD.               // Task 10
EEEEEEEEEEEEEEEE.               task10_ReverseArguments(new String[]{"Java", "Python",
   "C++", "JavaScript"});
FFFFFFFFFFFFFFFF.
GGGGGGGGGGGGGGGG.            } else {
HHHHHHHHHHHHHHHH.               // Run with actual command-line arguments
IIIIIIIIIIIIIIII.               System.out.println("Command-line arguments detected!");
JJJJJJJJJJJJJJJJ.               System.out.println("Running with provided
   arguments...\n");
KKKKKKKKKKKKKKKK.               System.out.println("=".repeat(60) + "\n");
LLLLLLLLLLLLLLLL.
MMMMMMMMMMMMMMMM.               // Run all tasks with actual arguments
NNNNNNNNNNNNNNNN.               task1_DisplayArguments(args);
OOOOOOOOOOOOOOOO.               task2_AddNumbers(args);
PPPPPPPPPPPPPPPP.               task3_MultiplyNumbers(args);
QQQQQQQQQQQQQQQQ.               task4_NameAndAge(args);
RRRRRRRRRRRRRRRR.               task5_CountArguments(args);
SSSSSSSSSSSSSSSS.               task6_AverageOfThree(args);
TTTTTTTTTTTTTTTT.               task7_HandleMissingArguments(args);
UUUUUUUUUUUUUUUU.               task8_FindMaximum(args);
VVVVVVVVVVVVVVVV.               task9_InterestCalculation(args);
WWWWWWWWWWWWWWWW.               task10_ReverseArguments(args);
XXXXXXXXXXXXXXXX.            }
YYYYYYYYYYYYYYYY.
ZZZZZZZZZZZZZZZZ.
AAAAAAAAAAAAAAAA.        }
BBBBBBBBBBBBBBBB.    }
CCCCCCCCCCCCCCCC.
DDDDDDDDDDDDDDDD.
```